Choice implies excluded middle by propositional suspension

The Univalent Foundations Program

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Editorial note. Construct and identify the path family of the suspension of a mere proposition, proving it is a set by double induction; splitting its Boolean surjection then produces decidability. The suspension construction is included, not granted as the conclusion.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained.

Source reference. Original reference 7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 166–166. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

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Source reference. Original reference 2.9.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1764–1764. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 3.6.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1807–1807. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 3.1.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1813–1813. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.7.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1853–1853. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 3.3.4: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1863–1863. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 4.7.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1891–1891. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 2.15.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1909–1909. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

We begin with the following lemma.

Lemma 10.0.1. *If A is a mere proposition then its suspension $\Sigma(A)$ is a set, and A is equivalent to $\mathsf{N} =_{\Sigma(A)} \mathsf{S}$.*

Proof. To show that $\Sigma(A)$ is a set, we define a family $P : \Sigma(A) \to \Sigma(A) \to \mathcal{U}$ with the property that $P(x, y)$ is a mere proposition for each $x, y : \Sigma(A)$, and which is equivalent to its identity type $\mathsf{Id}_{\Sigma(A)}$. We make the following definitions:

$$\begin{aligned} P(\mathsf{N}, \mathsf{N}) &:\equiv \mathbf{1} & \qquad P(\mathsf{S}, \mathsf{N}) &:\equiv A \\ P(\mathsf{N}, \mathsf{S}) &:\equiv A & \qquad P(\mathsf{S}, \mathsf{S}) &:\equiv \mathbf{1}. \end{aligned}$$

We have to check that the definition preserves paths. Given any $a : A$, there is a meridian $\mathsf{merid}(a) : \mathsf{N} = \mathsf{S}$, so we should also have

$$P(\mathsf{N}, \mathsf{N}) = P(\mathsf{N}, \mathsf{S}) = P(\mathsf{S}, \mathsf{N}) = P(\mathsf{S}, \mathsf{S}).$$

But since A is inhabited by a, it is equivalent to $\mathbf{1}$, so we have

$$P(\mathsf{N}, \mathsf{N}) \simeq P(\mathsf{N}, \mathsf{S}) \simeq P(\mathsf{S}, \mathsf{N}) \simeq P(\mathsf{S}, \mathsf{S}).$$

The univalence axiom turns these into the desired equalities. Also, $P(x, y)$ is a mere proposition for all $x, y : \Sigma(A)$, which is proved by induction on x and y, and using the fact that being a mere proposition is a mere proposition.

Note that P is a reflexive relation. Therefore we may apply Theorem 10.1.5, so it suffices to construct $\tau : \prod_{(x,y:\Sigma(A))} P(x, y) \to (x = y)$. We do this by a double induction. When x is N, we define $\tau(\mathsf{N})$ by

$$\tau(\mathsf{N}, \mathsf{N}, u) :\equiv \mathsf{refl}_{\mathsf{N}} \qquad \text{and} \qquad \tau(\mathsf{N}, \mathsf{S}, a) :\equiv \mathsf{merid}(a).$$

If A is inhabited by a then $\mathsf{merid}(a) : \mathsf{N} = \mathsf{S}$ so we also need $\mathsf{merid}(a)_*(\tau(\mathsf{N}, \mathsf{N})) = \tau(\mathsf{N}, \mathsf{S})$. This we get by function extensionality using the fact that, for all $x : A$,

$$\begin{aligned} \mathsf{merid}(a)_*(\tau(\mathsf{N}, \mathsf{N}, x)) = \tau(\mathsf{N}, \mathsf{N}, x) \cdot \mathsf{merid}(a)^{-1} \equiv \\ \mathsf{refl}_{\mathsf{N}} \cdot \mathsf{merid}(a) = \mathsf{merid}(a) = \mathsf{merid}(x) \equiv \tau(\mathsf{N}, \mathsf{S}, x). \end{aligned}$$

In a symmetric fashion we may define $\tau(\mathsf{S})$ by

$$\tau(\mathsf{S}, \mathsf{N}, a) :\equiv \mathsf{merid}(a)^{-1} \qquad \text{and} \qquad \tau(\mathsf{S}, \mathsf{S}, u) :\equiv \mathsf{refl}_{\mathsf{S}}.$$

To complete the construction of τ, we need to check $\mathsf{merid}(a)_*(\tau(\mathsf{N})) = \tau(\mathsf{S})$, given any $a : A$. The verification proceeds much along the same lines by induction on the second argument of τ.

Thus, by Theorem 10.1.5 we have that $\Sigma(A)$ is a set and that $P(x, y) \simeq (x = y)$ for all $x, y : \Sigma(A)$. Taking $x :\equiv \mathsf{N}$ and $y :\equiv \mathsf{S}$ yields $A \simeq (\mathsf{N} =_{\Sigma(A)} \mathsf{S})$ as desired. □

10.1 The axiom of choice

We can now properly formulate the axiom of choice in homotopy type theory. Assume a type X and type families

$$A : X \to \mathcal{U} \qquad \text{and} \qquad P : \prod_{x:X} A(x) \to \mathcal{U},$$

and moreover that

- X is a set,
- $A(x)$ is a set for all $x : X$, and
- $P(x, a)$ is a mere proposition for all $x : X$ and $a : A(x)$.

The **axiom of choice** AC asserts that under these assumptions,

$$\Big(\prod_{x:X} \Big\| \sum_{a:A(x)} P(x,a) \Big\|\Big) \to \Big\| \sum_{(g:\prod_{(x:X)} A(x))} \prod_{(x:X)} P(x, g(x)) \Big\|. \tag{10.1.1}$$

Of course, this is a direct translation of (3.2.1) where we read "there exists $x : A$ such that $B(x)$" as $\left\|\sum_{(x:A)} B(x)\right\|$, so we could have written the statement in the familiar logical notation as

$$\Big(\forall(x : X).\, \exists(a : A(x)).\, P(x,a)\Big) \Rightarrow \Big(\exists(g : \textstyle\prod_{(x:X)} A(x)).\, \forall(x : X).\, P(x, g(x))\Big).$$

In particular, note that the propositional truncation appears twice. The truncation in the domain means we assume that for every x there exists some $a : A(x)$ such that $P(x, a)$, but that these values are not chosen or specified in any known way. The truncation in the codomain means we conclude that there exists some function g, but this function is not determined or specified in any known way.

In fact, because of original reference 2.15.7, this axiom can also be expressed in a simpler form.

Lemma 10.1.2. *The axiom of choice* (10.1.1) *is equivalent to the statement that for any set X and any $Y : X \to \mathcal{U}$ such that each $Y(x)$ is a set, we have*

$$\Big(\prod_{x:X} \Big\| Y(x) \Big\|\Big) \to \Big\| \prod_{x:X} Y(x) \Big\|. \tag{10.1.3}$$

This corresponds to a well-known equivalent form of the classical axiom of choice, namely "the cartesian product of a family of nonempty sets is nonempty".

Proof. By original reference 2.15.7, the codomain of (10.1.1) is equivalent to

$$\Big\| \prod_{(x:X)} \sum_{(a:A(x))} P(x,a) \Big\|.$$

Thus, (10.1.1) is equivalent to the instance of (10.1.3) where $Y(x) :\equiv \sum_{(a:A(x))} P(x, a)$. (This is a set by original references 3.1.5 and 3.3.4.) Conversely, (10.1.3) is equivalent to the instance of (10.1.1) where $A(x) :\equiv Y(x)$ and $P(x, a) :\equiv \mathbf{1}$. Thus, the two are logically equivalent. Since both are mere propositions, by original reference 3.3.3 they are equivalent types. □

As with LEM, the equivalent forms (10.1.1) and (10.1.3) are not a consequence of our basic type theory, but they may consistently be assumed as axioms.

Remark 10.1.4. It is easy to show that the right side of (10.1.3) always implies the left. Since both are mere propositions, by original reference 3.3.3 the axiom of choice is also equivalent to asking for an equivalence

$$\Big(\prod_{x:X} \Big\| Y(x) \Big\|\Big) \simeq \Big\| \prod_{x:X} Y(x) \Big\|$$

This illustrates a common pitfall: although dependent function types preserve mere propositions (original reference 3.6.2), they do not commute with truncation: $\left\|\prod_{(x:A)} P(x)\right\|$ is not generally equivalent to $\prod_{(x:A)}\|P(x)\|$. The axiom of choice, if we assume it, says that this is true *for sets*; as we will see below, it fails in general.

The restriction in the axiom of choice to types that are sets can be relaxed to a certain extent. For instance, we may allow A and P in (10.1.1), or Y in (10.1.3), to be arbitrary type families; this results in a seemingly stronger statement that is equally consistent. We may also replace the propositional truncation by the more general n-truncations to be considered in original reference 7, obtaining a spectrum of axioms AC_n interpolating between (10.1.1), which we call simply AC (or AC_{-1} for emphasis), and original reference 2.15.7, which we shall call AC_∞. See also original references 7.8 and 7.10.

Theorem 10.1.5. *Suppose R is a reflexive mere relation on a type X implying identity. Then X is a set, and $R(x,y)$ is equivalent to $x =_X y$ for all $x, y : X$.*

Proof. Let $\rho : \prod_{(x:X)} R(x,x)$ witness reflexivity of R, and let $f : \prod_{(x,y:X)} R(x,y) \to (x =_X y)$ be a witness that R implies identity. Note first that the two statements in the theorem are equivalent. For on one hand, if X is a set, then $x =_X y$ is a mere proposition, and since it is logically equivalent to the mere proposition $R(x,y)$ by hypothesis, it must also be equivalent to it. On the other hand, if $x =_X y$ is equivalent to $R(x,y)$, then like the latter it is a mere proposition for all $x, y : X$, and hence X is a set.

We give two proofs of this theorem. The first shows directly that X is a set; the second shows directly that $R(x,y) \simeq (x = y)$.

First proof: we show that X is a set. The idea is the same as that of original reference 3.3.4: the function f must be continuous in its arguments x and y. However, it is slightly more notationally complicated because we have to deal with the additional argument of type $R(x,y)$.

Firstly, for any $x : X$ and $p : x =_X x$, consider $\mathsf{apd}_{f(x)}(p)$. This is a dependent path from $f(x,x)$ to itself. Since $f(x,x)$ is still a function $R(x,x) \to (x =_X x)$, by original reference 2.9.6 this yields for any $r : R(x,x)$ a path

$$p_*(f(x,x,r)) = f(x,x,p_*(r)).$$

On the left-hand side, we have transport in an identity type, which is concatenation. And on the right-hand side, we have $p_*(r) = r$, since both lie in the mere proposition $R(x,x)$. Thus, substituting $r :\equiv \rho(x)$, we obtain

$$f(x,x,\rho(x)) \cdot p = f(x,x,\rho(x)).$$

By cancellation, $p = \mathsf{refl}_x$. So X satisfies Axiom K, and hence is a set.

Second proof: we show that each $f(x,y) : R(x,y) \to x =_X y$ is an equivalence. By original reference 4.7.7, it suffices to show that f induces an equivalence of total spaces:

$$\Big(\sum_{y:X} R(x,y)\Big) \simeq \Big(\sum_{y:X} x =_X y\Big).$$

By original reference 3.11.8, the type on the right is contractible, so it suffices to show that the type on the left is contractible. As the center of contraction we

take the pair $(x, \rho(x))$. It remains to show, for every $y : X$ and every $H : R(x, y)$ that

$$(x, \rho(x)) = (y, H).$$

But since $R(x, y)$ is a mere proposition, by original reference 2.7.2 it suffices to show that $x =_X y$, which we get from $f(H)$. □

Theorem 10.1.6 (Diaconescu)**.** *The axiom of choice implies the law of excluded middle.*

Proof. We use the equivalent form of choice given in Lemma 10.1.2. Consider a mere proposition A. The function $f : \mathbf{2} \to \Sigma(A)$ defined by $f(0_{\mathbf{2}}) :\equiv \mathsf{N}$ and $f(1_{\mathbf{2}}) :\equiv \mathsf{S}$ is surjective. Indeed, we have $(0_{\mathbf{2}}, \mathsf{refl}_{\mathsf{N}}) : \mathsf{fib}_f(\mathsf{N})$ and $(1_{\mathbf{2}}, \mathsf{refl}_{\mathsf{S}}) : \mathsf{fib}_f(\mathsf{S})$. Since $\left\|\mathsf{fib}_f(x)\right\|$ is a mere proposition, by induction the claimed surjectivity follows.

By Lemma 10.0.1 the suspension $\Sigma(A)$ is a set, so by the axiom of choice there merely exists a section $g : \Sigma(A) \to \mathbf{2}$ of f. As equality on $\mathbf{2}$ is decidable we get

$$(g(f(0_{\mathbf{2}})) = g(f(1_{\mathbf{2}}))) + \neg(g(f(0_{\mathbf{2}})) = g(f(1_{\mathbf{2}}))),$$

and, since g is a section of f, hence injective,

$$(f(0_{\mathbf{2}}) = f(1_{\mathbf{2}})) + \neg(f(0_{\mathbf{2}}) = f(1_{\mathbf{2}})).$$

Finally, since $(f(0_{\mathbf{2}}) = f(1_{\mathbf{2}})) = (\mathsf{N} = \mathsf{S}) = A$ by Lemma 10.0.1, we have $A + \neg A$. □